

Report of Calibration
PSC 2CH-HSS-AR-QD-QF_S/N 0020
2025-12-03 19:30:16

Calibration Current standard: BNL PSC ATE S/N 001
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647
Calibration Resistance standard: Fluke 742A-1 S/N 1063008
End Header

lab{4}Chan1
Burden resistor = 18.0000

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.001006	0.993137	-1.023846	-1.026933	0.998004	0.030773
-50.006451	50.020772	-49.660183	-49.682499	50.142166	0.067817

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.008322	-0.030379	-0.033074	0.000000
Initial measured gains:	1.000453	0.992468	0.992860	1.002377
Gain corrections:	1.000453	1.007589	1.007191	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	0.997039
50.000000	50.135635

Measured offset: -0.005789
Measured gain: 1.002828
Gain correction: 0.997179

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.001079	1.000746	-1.001111	-1.001118	1.001273	0.025611
-50.006694	50.005600	-50.005947	-50.005997	50.005516	0.054259

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000317	-0.000048	-0.000054	0.000539
Final measured gains:	0.999984	0.999984	0.999985	0.999988

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000093	-0.000051	-0.000026	0.000446
Final measured offsets stdev:	0.000154	0.000121	0.000081	0.000071
Final measured gains mean:	0.999978	0.999978	0.999981	0.999994
Final measured gains stdev:	0.000006	0.000004	0.000004	0.000009

Saving channel 1 calibration constants to qspi

lab{4}Chan2
Burden resistor = 9.0000

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000710	0.989748	-1.063250	-1.056537	1.007121	0.023764
-100.012960	100.080430	-99.368599	-99.323334	100.317955	0.143222

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.011754	-0.069685	-0.063361	0.000000
Initial measured gains:	1.000792	0.992860	0.992471	1.002222
Gain corrections:	1.000792	1.007191	1.007586	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.006552
100.000000	100.304840

Measured offset: 0.003539
Measured gain: 1.003013
Gain correction: 0.996996

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.000802	1.000407	-1.000885	-1.001011	1.003730	0.020365
-100.013418	100.011404	-100.012131	-100.012375	100.014771	0.137947

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000379	-0.000096	-0.000222	0.003323
Final measured gains:	0.999984	0.999986	0.999987	1.000000

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000371	-0.000024	-0.000070	0.003395
Final measured offsets stdev:	0.000187	0.000058	0.000122	0.000324
Final measured gains mean:	0.999981	0.999983	0.999987	0.999991
Final measured gains stdev:	0.000007	0.000007	0.000003	0.000007

Saving channel 2 calibration constants to qspi

Test data reviewed by  Date 12/5/25